

ALPINE, TX, USA

WMO: 720151

Lat: 30.383N

Lon: 103.683W

Elev: 1376

StdP: 85.85

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03049

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.5	(c) -4.5	(d) -19.0	(e) 0.8	(f) 11.2	(g) -16.2	(h) 1.1	(i) 12.2	(j) 13.3	(k) 13.2	(l) 12.1	(m) 13.8	(n) 2.4	(o) 50	(p) 0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 14.4	(c) 36.0	(d) 16.9	(e) 34.9	(f) 16.9	(g) 33.4	(h) 16.9	(i) 20.2	(j) 27.6	(k) 19.8	(l) 27.5	(m) 19.4	(n) 27.2	(o) 4.2	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	15.8	21.8	17.9	15.2	21.8	17.4	14.8	21.7	64.8	27.7	63.0	27.4	61.6	27.1	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.1	(b) 9.2	(c) 8.1	(e) -10.1	(f) 37.9	(g) 2.7	(h) 1.2	(i) -12.1	(j) 38.8	(k) -13.7	(l) 39.5	(m) -15.2	(n) 40.1	(o) -17.2	(p) 41.0		
(a) 11.1	(b) 9.2	(c) 8.1	(e) -10.6	(f) 21.4	(g) 2.5	(h) 0.4	(i) -12.4	(j) 21.6	(k) -13.9	(l) 21.8	(m) -15.3	(n) 22.0	(o) -17.1	(p) 22.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	7.5	10.3	13.9	17.7	21.7	25.7	25.1	24.6	21.5	17.5	11.8	8.4
	DBStd	7.68	5.11	5.70	4.95	4.12	4.06	2.79	2.44	2.61	3.07	4.35	4.98	5.26
	HDD10.0	333	111	60	22	2	1	0	0	0	0	6	39	93
	HDD18.3	1369	337	228	147	59	17	0	1	0	9	65	199	308
	CDD10.0	2956	32	68	144	233	364	472	469	453	346	239	94	43
	CDD18.3	951	0	2	11	41	122	222	211	194	105	40	3	0
	CDH23.3	10833	4	64	248	674	1592	2648	2169	1924	958	479	66	6
	CDH26.7	4724	0	9	42	205	736	1398	1006	868	334	120	6	0
Wind	WSAvg	3.6	3.9	4.2	4.1	4.3	4.0	3.5	3.0	2.7	2.8	3.2	3.6	3.9
Precipitation	PrecAvg	409	12	13	9	13	28	58	75	75	72	37	15	15
	PrecMax	698	41	75	43	92	89	178	238	208	283	113	82	66
	PrecMin	203	0	0	0	0	0	0	1	11	1	0	0	0
	PrecStd	116	11	17	11	19	22	41	54	43	59	33	17	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	28.0	29.8	32.7	36.6	38.1	37.0	35.9	33.9	31.4	27.5	24.2
		MCWB	10.3	11.7	12.2	13.2	15.0	15.9	17.6	18.1	17.2	15.8	12.7	10.4
	2%	DB	21.5	25.3	27.8	30.8	34.5	36.6	34.9	34.5	32.2	29.7	25.2	22.3
		MCWB	9.2	10.5	11.3	12.2	14.6	16.5	17.8	17.9	16.7	15.3	12.3	9.9
	5%	DB	19.3	22.9	26.2	29.0	32.9	35.2	33.2	32.9	30.4	27.9	23.3	20.7
		MCWB	8.1	9.5	11.0	11.8	14.1	16.5	17.8	17.8	17.0	15.2	11.5	9.0
	10%	DB	16.9	21.1	24.5	27.4	31.3	33.6	32.3	31.9	28.5	26.2	21.3	17.7
		MCWB	6.9	8.8	10.1	11.5	13.9	17.0	18.0	18.1	16.7	14.7	10.6	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	13.2	15.3	16.6	19.8	20.8	20.7	20.7	20.4	19.6	15.3	12.2
		MCDB	20.6	23.9	22.5	24.2	26.8	28.3	28.2	27.4	26.5	26.7	21.2	20.9
	2%	WB	10.3	11.6	13.5	15.4	18.6	20.1	20.1	20.0	19.8	18.5	14.1	10.9
		MCDB	19.2	22.2	22.2	23.0	25.0	28.2	27.8	27.6	26.3	24.6	20.2	19.9
	5%	WB	8.9	10.5	12.4	14.4	17.8	19.5	19.6	19.5	19.3	17.5	12.9	9.8
		MCDB	18.1	21.1	21.9	22.9	24.7	28.0	27.5	27.4	25.6	23.6	20.7	18.6
	10%	WB	7.6	9.4	11.2	13.3	16.9	19.0	19.1	19.1	18.7	16.5	11.8	8.6
		MCDB	16.2	19.6	21.6	23.0	24.6	27.7	27.3	27.0	24.8	22.9	20.0	16.7
Mean Daily Temperature Range	5% DB	MDBR	14.2	15.5	15.9	16.2	15.3	14.4	12.8	12.6	12.4	14.3	14.9	14.3
		MCDBR	16.0	17.2	17.4	17.5	17.1	15.7	14.6	14.4	15.1	15.4	16.4	16.2
		MCWBR	7.9	7.7	7.4	7.2	6.2	4.7	3.9	3.6	4.5	5.6	7.3	7.9
	5% WB	MCDBR	15.2	16.0	15.3	15.7	15.3	14.2	12.5	12.0	11.3	12.5	14.5	15.3
		MCWBR	7.8	7.7	6.8	6.7	6.6	4.8	3.7	3.8	3.9	5.1	7.2	8.0
Clear-Sky Solar Irradiance	taub	0.249	0.256	0.269	0.300	0.317	0.347	0.369	0.349	0.331	0.288	0.265	0.251	
	taud	2.676	2.643	2.595	2.489	2.481	2.455	2.433	2.513	2.510	2.636	2.632	2.656	
	Ebn at Noon	1010	1025	1023	992	969	933	912	931	942	974	979	990	
	Edh at Noon	74	84	94	109	111	113	115	105	102	83	76	72	
All-Sky Solar Radiation	RadAvg	3.73	4.70	5.92	7.01	7.41	7.34	6.74	6.33	5.58	4.86	3.96	3.45	
	RadStd	0.34	0.41	0.33	0.40	0.40	0.35	0.39	0.40	0.35	0.44	0.34	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	-174	+330	+213
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103

Nomenclature: See separate page